AUTONOMOUS SYSTEMS BLUEPRINT • V3.8

$49 ARCHITECTURE VAULT

SOLOPRENEUR AI AUTOMATION & MULTI-AGENT WORKFLOWS PLAYBOOK

Step-by-Step Production Architectures for Autonomous Content Pipelines, 60-Second Lead Qualification Bots, Zero-Touch Stripe/CRM Webhooks & Multi-Agent Executive Teams (n8n, Make.com & OpenAI/Claude APIs)

WHAT IS INSIDE THIS EXECUTIVE SYSTEM VAULT:

- **01. Autonomous Omnichannel Content Engine (1 Idea -> 15 Assets)**
  End-to-end Make/n8n pipeline turning voice notes or industry news into SEO articles, LinkedIn carousels, X threads, and newsletter drafts.
- **02. Sub-60-Second AI Lead Qualification, Enrichment & Routing Bot**
  Webhook-driven Clearbit/Apollo enrichment, LLM ICP scoring, personalized draft generation, and instant Slack/Calendly fast-lane routing.
- **03. Zero-Touch Finance, Stripe Invoicing, Onboarding & Churn Recovery**
  Automated contract generation, client workspace provisioning, dunning payment recovery, and AI executive weekly financial digests.
- **04. Multi-Agent Supervisor-Worker Teams (LangGraph / CrewAI / n8n)**
  Orchestrator-Router-Specialist-Critic patterns with deterministic JSON schemas, tool calling, and human-in-the-loop approval gates.
- **05. Production Reliability, Token Cost FinOps & Guardrail Engineering**
  Idempotency keys, exponential backoff, prompt caching (-75% API cost), and hallucination prevention schemas.

25+ HRS/WK
MANUAL BUSYWORK ELIMINATED

12 BLUEPRINTS
NODE-BY-NODE SYSTEM ARCHITECTURES

99.8% UPTIME
VOID ERROR-HANDLING & FALLBACKS

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Every module in this guidebook is engineered for immediate deployment. Use the copy-paste frameworks, system prompts, and architectural blueprints in sequence or jump directly to your high-leverage bottleneck.

MODULE 01: The Autonomous Solopreneur Operating System & Stack Selection — Module 01
n8n vs. Make.com vs. Zapier vs. Code, API token economics, and the 4-Tier Automation Maturity Model.

MODULE 02: Automated Omnichannel Content Engine (Research -> Multi-Channel) — Module 02
Perplexity research ingestion, brand-voice synthesis, visual generation, and human-in-the-loop one-click publishing.

MODULE 03: AI Lead Qualification, Enrichment & Instant Speed-to-Lead Bots — Module 03
Webhook capture, Clay/Apollo scraping, deterministic ICP JSON scoring, and automated CRM/Slack routing.

MODULE 04: Zero-Touch Client Onboarding, Invoicing & CRM Webhook Pipelines — Module 04
Stripe checkout triggers, Notion/ClickUp portal provisioning, AI kickoff briefs, and failed-payment recovery.

MODULE 05: Multi-Agent Team Architectures (Supervisor, Researcher, Critic) — Module 05
Designing collaborative multi-agent swarms in n8n and LangGraph with shared memory and strict quality gates.

MODULE 06: Production Reliability, FinOps Cost Control & JSON Schemas — Module 06
Strict JSON mode, dead-letter queues, prompt caching, rate-limit handling, and the 7-Day Deployment Checklist.

[PRO TIP] HOW TO USE THIS VAULT FOR MAXIMUM ROI

1. Bracketed variables like [TARGET AUDIENCE], [OFFER PRICE], or [TECH STACK] are dynamic slots—replace them with your specific context before running.
2. Chain prompts sequentially: use the output of the Research/Architecture prompt as the context window input for the Execution prompt.
3. Store your top-performing customized prompts in a dedicated Notion, Obsidian, or Custom GPT / Claude Project knowledge base.

MODULE 01 • IMPLEMENTATION BLUEPRINT

The Autonomous Solopreneur Stack & Architecture Principles

How a 1-person business operates with the output capacity of a 12-person agency by replacing manual glue-work with deterministic webhook & agent pipelines.

OPERATING LEVERAGE: $500k-$2M Revenue Capacity Per Founder

Most solopreneurs trap themselves in "fragile automation"—stitching together 15 single-step Zaps that break silently whenever an LLM returns an extra conversational sentence. Production AI automation requires treating LLMs not as chat partners, but as deterministic JSON transformation microservices inside a state-managed pipeline.

Orchestration Layer	Primary Tooling (2026)	Monthly Cost	Best Strategic Use Case
Visual Workflow & AI Agents	n8n (Self-Hosted on Hetzner/ Railway)	$6 - $20/mo	Unlimited complex multi-step workflows, native LangChain agent nodes, webhooks, zero per-task tax.
Rapid Cloud Orchestration	Make.com (Core / Pro Plan)	$10 - $29/mo	Visual HTTP routers, instant OAuth integrations with 1,500+ SaaS apps, JSON iterators/aggregators.
Data & Outbound Enrichment	Clay.com + Apollo + Perplexity API	$49 - $149/mo	Waterfall lead enrichment, live web scraping, and programmatic outbound personalization.
Database & State Backend	Supabase (PostgreSQL) / Airtable	$0 - $25/mo	Storing execution logs, idempotency keys, content queues, and vector embeddings (pgvector).
LLM Intelligence Routing	Claude 3.5 Sonnet + GPT-4o-mini	$15 - $60/mo API	GPT-4o-mini for cheap classification/extraction; Claude 3.5 Sonnet for elite copy & code synthesis.

[ARCHITECTURAL INVARIANT] THE GOLDEN RULE OF AI AUTOMATION: DETERMINISTIC CODE AROUND PROBABILISTIC AI

Never ask an LLM to format dates, calculate invoice taxes, or decide which API endpoint URL to call in free-text. Use deterministic code/nodes for routing, math, and API calls—and restrict the LLM strictly to classification, summarization, extraction, and creative drafting using `response_format: { type: "json_object" }`.

MODULE 02 • IMPLEMENTATION BLUEPRINT

Blueprint #1: The Autonomous Omnichannel Content Engine

Turn a single 3-minute voice memo, YouTube URL, or industry news trigger into 12 platform-native assets stored in Notion/Airtable ready for 1-click approval.

TIME SAVED: 14 Hours/Week | COST PER RUN: $0.08 in API Tokens

ARCHITECTURE BLUEPRINT: OMNICHANNEL CONTENT REPURPOSING & PUBLISHING PIPELINE

Stack: n8n / Make.com + Whisper + Perplexity API + Claude 3.5 Sonnet + Notion + Buffer/Typefully | Target ROI: 15+ High-Authority Posts/Week with 20 Mins Review

NODE 01

TRIGGER & INGEST: Webhook / Slack Voice Note / RSS Feed Monitor

Watches a Slack channel, Notion database status change, or Perplexity/RSS feed. Transcribes audio via OpenAI Whisper API.

NODE 02

RESEARCH ENRICHMENT: Perplexity Sonar Pro API Deep Fact-Check

Queries live web citations, recent statistics, and contrarian industry angles so the content contains real Information Gain.

NODE 03

CORE SYNTHESIS: Claude 3.5 Sonnet Master Pillar Generator

Synthesizes a 1,800-word authoritative newsletter / SEO article injected with your Few-Shot Brand Voice JSON file.

NODE 04

PARALLEL SPLITTER: Platform-Specific Derivative Agents (JSON Output)

Generates 3 LinkedIn posts, 1 8-tweet X thread, 2 Short-form 60s video scripts, and 1 carousel slide deck in one structured JSON payload.

NODE 05

APPROVAL & SCHEDULE: Airtable / Notion Staging + Slack Interactive Buttons

Pushes drafts to Notion and sends a Slack preview with [Approve & Schedule] or [Regenerate with Feedback] webhook buttons.

ERROR HANDLING & FALLBACK: Enable HTTP 429 exponential backoff + dead-letter Slack alert webhook on every LLM node.

PROMPT #A201 System Prompt: Omnichannel Content Repurposing JSON Engine — Claude 3.5 Sonnet (API JSON Mode)

```
SYSTEM ROLE: You are an Autonomous Content Repurposing Engine running inside an API
pipeline.
BRAND VOICE PROFILE: [CONCISE, PRACTITIONER-LED, ZERO FLUFF, QUANTIFIED EXAMPLES, NO
EMOJIS OR HASHTAGS]
INPUT SOURCE TRANSCRIPT & RESEARCH: {{source_text}} | {{perplexity_citations}}
TASK: Transform the input into a complete omnichannel content package and return ONLY
valid JSON matching the exact schema below:
{
  "core_thesis": "1-sentence contrarian takeaway",
  "newsletter_issue": {
    "subject_lines": ["Option 1", "Option 2", "Option 3"],
    "preview_text": "Under 90 chars",
    "markdown_body": "Full 900-word practitioner essay with H2 headers, bulleted
framework, and action step"
  },
  "linkedin_posts": [
    { "angle": "Framework Breakdown", "hook": "First 2 lines before see-more fold",
"body": "Full post under 1,300 chars" },
    { "angle": "Contrarian Myth-Buster", "hook": "First 2 lines", "body": "Full post" }
  ],
  "x_thread": ["Tweet 1 (Hook + Proof)", "Tweet 2", "Tweet 3", "Tweet 4", "Tweet 5
(Summary + CTA)"],
  "short_video_script_60s": { "hook_0_3s": "...", "body_3_45s": "...", "cta_45_60s":
"..." }
}
STRICT CONSTRAINT: Output raw JSON only. Never wrap in markdown code fences if parsed
directly by HTTP module.
```

EXPECTED DELIVERABLE: Deterministic JSON payload containing newsletter, 2 LinkedIn posts, 5-part X thread, and 60s video script.

OPTIMIZATION LEVER: Map each JSON key directly to Airtable/Notion columns using Make.com Parse JSON module.

PROMPT #A202 System Prompt: Perplexity Live Research & Citation Enricher — Perplexity Sonar-Reasoning-Pro API

```
ROLE: You are a Real-Time Investigative Research Node.
TOPIC / DRAFT THESIS: {{raw_idea}}
TASK: Find 4 concrete, verifiable 2025-2026 data points, case studies, or technical
benchmarks that validate or challenge this thesis.
OUTPUT SCHEMA:
1. Key Statistic #1 - #4 (Include exact number, source organization, and publication
context).
2. Common Industry Misconception: What do 90% of blog posts get wrong about this topic?
3. Practitioner Nuance: One technical or operational edge case that only experienced
operators know.
```

EXPECTED DELIVERABLE: Live web-grounded facts and benchmarks fed into the Claude writer node to eliminate generic AI output.

OPTIMIZATION LEVER: Always chain Perplexity BEFORE Claude—Perplexity searches the live web while Claude writes superior prose.

MODULE 03 • IMPLEMENTATION BLUEPRINT

Blueprint #2: Sub-60-Second AI Lead Qualification & Speed-to-Lead Engine

Contacting an inbound lead within 60 seconds increases conversion by 391%. Here is how to enrich, score, and respond to every lead automatically.

SPEED TO LEAD: 18 Seconds Average | PIPELINE CONVERSION: +210%

ARCHITECTURE BLUEPRINT: INSTANT INBOUND LEAD ENRICHMENT, SCORING & PERSONALIZED RESPONSE BOT

Stack: n8n / Make + Jina Reader API + Apollo API + GPT-4o-mini + Claude 3.5 Sonnet + HubSpot + Slack | Target ROI: 3.9x More Booked Discovery Calls from Same Traffic

NODE 01

WEBHOOK CAPTURE: Typeform / Tally / Website Form / Calendly Webhook

Captures Name, Work Email, Website URL, and free-text Project Description in real time.

NODE 02

DOMAIN & LINKEDIN SCRAPE: HTTP Scraper + Apollo/Clearbit API Enrichment

Extracts the prospect company homepage markdown (via Jina Reader r.jina.ai/{{domain}}), employee count, funding, and tech stack.

NODE 03

AI ICP SCORING: GPT-4o-mini Structured Lead Classifier (0-100 Score)

Evaluates fit against ICP criteria, categorizes budget tier (Tier 1 VIP / Tier 2 Self-Serve / Tier 3 Disqualified), and extracts pain points.

NODE 04

TAILORED RESPONSE: Claude 3.5 Sonnet Hyper-Personalized Audit & Reply Draft

Generates a bespoke 3-point mini-audit of their website + custom reply email referencing their exact tech stack and bottleneck.

NODE 05

MULTI-PATH ROUTER: HubSpot/Pipedrive CRM + Slack VIP Alert + Gmail Draft

Tier 1 leads trigger an instant Slack alert + immediate personalized email with priority booking link; Tier 2 enters self-serve sequence.

ERROR HANDLING & FALLBACK: Enable HTTP 429 exponential backoff + dead-letter Slack alert webhook on every LLM node.

PROMPT #A301 **System Prompt: Deterministic B2B Lead Scoring & Routing Classifier** GPT-4o-mini (Structured Outputs JSON)

```
ROLE: You are an Autonomous RevOps Lead Qualification Classifier.
IDEAL CUSTOMER PROFILE (ICP) CRITERIA:
- Target Industries: [B2B SaaS, E-Commerce Brands >$50k/mo, Digital Agencies]
- Minimum Budget / Revenue Signal: [$3,000+ project or 5+ employees]
- Disqualifiers: [Students, free gmail/yahoo addresses with no website, competitors]
LEAD PAYLOAD:
- Name: {{name}} | Email: {{email}} | Website: {{website}}
- Stated Challenge: {{form_message}}
- Enriched Company Data & Homepage Scrape: {{company_scrape}}
TASK: Score and classify this lead into strict JSON:
{
  "icp_score": 85,
  "routing_tier": "TIER_1_VIP | TIER_2_NURTURE | TIER_3_DISQUALIFIED",
  "detected_company_summary": "1 sentence on what their business actually sells",
  "primary_pain_vector": "Core bottleneck identified from form + website",
  "custom_audit_hook": "1 specific observation from their website we can mention in our
reply",
  "recommended_offer": "Which of our packages fits best"
}
```

EXPECTED DELIVERABLE: JSON classification payload used by Make/n8n Switch Router to branch Tier 1 vs Tier 2 leads.

OPTIMIZATION LEVER: Use `https://r.jina.ai/https://theirwebsite.com` in an HTTP GET node to convert any prospect website into clean LLM-readable markdown for free.

PROMPT #A302 **System Prompt: Instant Speed-to-Lead Bespoke Reply Generator** Claude 3.5 Sonnet

```
ROLE: Act as the Founder responding personally to a high-intent inbound inquiry within 2
minutes of submission.
LEAD CONTEXT: {{lead_json_from_previous_node}}
TASK: Write a warm, sharp, 110-word plain-text email reply that proves we actually looked
at their business.
STRUCTURE:
1. Greeting + specific observation from their live site ({{custom_audit_hook}}).
2. 2 concrete bullets on how we would solve "{{primary_pain_vector}}" based on a similar
system we built.
3. Direct link to book our Priority Architecture Call (or offer to send a 90-second Loom
video if they prefer async).
CONSTRAINTS: Zero corporate filler ("Thank you for reaching out to us"). Sound like a
busy, sharp operator who noticed their note right away.
```

EXPECTED DELIVERABLE: Human-grade personalized reply sent automatically 90 seconds after form submission.

OPTIMIZATION LEVER: Add a 90-second delay node before sending so the email feels naturally typed by a human rather than an instant auto-responder.

MODULE 04 • IMPLEMENTATION BLUEPRINT

Blueprint #3: Zero-Touch Client Onboarding, Invoicing & Churn Recovery

Eliminate the 2-hour manual chaos after a client pays. Automate contract generation, workspace creation, kickoff roadmaps, and failed-payment dunning.

ONBOARDING SPEED: 12 Seconds After Stripe Payment | CHURN RECOVERY: 64% Saved

ARCHITECTURE BLUEPRINT: STRIPE PAYMENT -> CLIENT PORTAL PROVISIONING & AI KICKOFF ARCHITECT

Stack: Stripe Webhooks + n8n/Make + Notion API + Google Drive API + Slack API + Claude 3.5 Sonnet | Target ROI: 100% Zero-Touch Onboarding & Immediate Buyer Confidence

NODE 01

PAYMENT WEBHOOK: Stripe checkout.session.completed / invoice.paid Event

Verifies Stripe HMAC signature, checks Supabase idempotency table to prevent duplicate execution, extracts customer metadata.

NODE 02

CRM & ACCOUNTING: HubSpot Deal Closed-Won + QuickBooks/Xero Receipt Sync

Updates deal stage, logs net revenue after Stripe fees, and generates compliant PDF tax invoice.

NODE 03

PORTAL PROVISIONING: Notion Client Portal + Google Drive Folder + Slack Connect

Clones master Notion Client Portal template, creates dedicated Google Drive folder with permissions, and invites client to private Slack channel.

NODE 04

AI KICKOFF ROADMAP: Claude 3.5 Sonnet 30-Day Custom Execution Plan Builder

Reads sales call transcript + intake notes from CRM and populates the client Notion portal with a custom 30-day milestone plan.

NODE 05

WELCOME DISPATCH: Personalized Welcome Email + Slack Kickoff Summary

Sends the client a single unified email containing their Portal Link, Intake Form, and Kickoff Call calendar link.

ERROR HANDLING & FALLBACK: Enable HTTP 429 exponential backoff + dead-letter Slack alert webhook on every LLM node.

PROMPT #A401 System Prompt: Automated Client Kickoff Roadmap & Portal Populator — Claude 3.5 Sonnet

```
ROLE: Act as a Senior Technical Project Director generating a client-facing 30-Day
Implementation Roadmap inside their new Notion Portal.
CLIENT DATA:
- Company: {{company_name}} | Package Purchased: {{package_name}}
- Discovery Call Notes / Goals: {{crm_notes}}
TASK: Generate the complete Markdown content for their Client Portal Welcome Page.
SECTIONS TO GENERATE:
1. Executive North Star: Restate their #1 quantified 30-day and 90-day success metrics.
2. Week-by-Week Sprint Schedule (Weeks 1 to 4): Clear deliverables, owner (Us vs. Client),
and target completion dates starting from {{today_date}}.
3. "What We Need From You in the Next 48 Hours": A 4-item checklist of credentials,
assets, or intake answers needed to prevent project delays.
4. Communication Protocol: How to use our shared Slack channel, async Loom updates, and
weekly sprint reviews.
```

EXPECTED DELIVERABLE: Custom-tailored Notion Client Portal home page populated automatically upon payment.

OPTIMIZATION LEVER: Experiencing a customized portal 30 seconds after paying eliminates buyer's remorse and cuts refund requests to near zero.

PROMPT #A402 **System Prompt: Smart Failed-Payment (Dunning) & Involuntary Churn Recovery** GPT-4o / Claude 3.5 Sonnet

```
ROLE: Act as a Customer Success & Retention Specialist handling a Stripe
`invoice.payment_failed` webhook event.
CUSTOMER CONTEXT:
- Name: {{first_name}} | Plan: {{plan_name}} | Tenure: {{months_active}} months
- Failure Decline Code: {{stripe_decline_code}} (e.g., insufficient_funds, expired_card,
generic_decline)
- Recent Product Usage / Value Delivered: {{usage_stats}}
TASK: Generate a 3-Step Empathetic Dunning Sequence (Day 0, Day 3, Day 6) + 1 SMS reminder
that recovers the card update without sounding like an aggressive debt collector.
KEY PSYCHOLOGY:
1. Tailor the wording to the decline code (e.g., if expired_card: "Looks like your card on
file recently expired").
2. Remind them of the live automations/assets currently running in their account
({{usage_stats}}) so they feel loss aversion if service pauses.
3. Provide a single frictionless pre-authenticated Stripe Billing Portal magic link.
```

EXPECTED DELIVERABLE: Context-aware 3-email + SMS payment recovery sequence triggered by Stripe webhook.

OPTIMIZATION LEVER: Recovering just 60% of involuntary failed cards adds 8-12% pure net profit to any subscription business.

MODULE 05 • IMPLEMENTATION BLUEPRINT

Blueprint #4: Multi-Agent Supervisor, Specialist & Critic Swarms

Why single mega-prompts hit a ceiling on complex tasks—and how to wire Supervisor-Worker-Critic multi-agent loops in n8n, LangGraph, or CrewAI.

QUALITY LIFT: +280% Accuracy via Adversarial Critic Loop

When a single LLM is asked to research, write, fact-check, and format a complex deliverable simultaneously, attention dilution causes it to cut corners. Multi-Agent Architecture splits cognitive roles across specialized narrow agents governed by a deterministic Supervisor and an adversarial Critic node.

Agent Node Role	Recommended Model	Temperature	Core Responsibility & Tool Access
1. Supervisor / Router	GPT-4o / Claude Sonnet	0.1	Deconstructs user goal into a JSON task DAG, assigns specialist agents, tracks state, and merges final output.
2. Deep Researcher	Perplexity Sonar Pro / Gemini	0.2	Executes web search, scrapes URLs via Jina/ Firecrawl, queries SQL/vector DB, and outputs cited evidence packets.
3. Domain Specialist	Claude 3.5 Sonnet	0.5	Synthesizes the core asset (copy, code, financial model) strictly from the Researcher evidence packet.
4. Adversarial Critic	OpenAI o3-mini / Sonnet	0.1	Scores the draft against a 10-point rubric. If score < 85/100, returns specific line-item fixes back to Specialist (max 2 loops).
5. Schema Formatter	GPT-4o-mini	0.0	Validates final output against strict JSON Schema and dispatches payloads to downstream APIs (Slack, Notion, GitHub).

ARCHITECTURE BLUEPRINT: AUTONOMOUS COMPETITOR INTELLIGENCE & WEEKLY EXECUTIVE BRIEFING SWARM

Stack: n8n AI Agent Nodes + Firecrawl + Supabase + Claude 3.5 Sonnet + OpenAI o3-mini + Slack | Target ROI: Full Competitive Intelligence Team for $4/Month in API Calls

NODE 01

CRON SCHEDULE: Monday 06:00 AM Cron Trigger + Competitor Watchlist DB

Loads top 5 competitor domains, pricing URLs, changelog pages, and Ad Library links from Supabase/ Airtable.

NODE 02

AGENT 1: SCRAPER: Firecrawl / Jina Differential Change Detector

Scrapes live pricing & feature pages, diffs against last week's snapshot hash, and isolates exact text/ price changes.

NODE 03

AGENT 2: ANALYST: Strategic Threat & Opportunity Synthesizer

Analyzes competitor moves, new ad angles, and customer complaints on G2/Reddit over the past 7 days.

NODE 04

AGENT 3: CRITIC: Hallucination & Actionability Gatekeeper

Verifies every claim has a source URL and rejects vague advice until 3 concrete counter-moves are specified.

NODE 05

DISPATCH: Executive PDF/Notion Brief + Slack Audio Summary

Publishes the Monday Morning Intelligence Brief to Notion and posts a 90-second ElevenLabs audio summary in Slack.

ERROR HANDLING & FALLBACK: Enable HTTP 429 exponential backoff + dead-letter Slack alert webhook on every LLM node.

PROMPT #A501 System Prompt: Multi-Agent Supervisor & Task Decomposition Router — GPT-4o / Claude 3.5 Sonnet

```
ROLE: You are the Supervisor Orchestrator of a 4-Agent Autonomous Execution Team:
[RESEARCHER, COPYWRITER, DATA_ANALYST, CRITIC].
INCOMING OBJECTIVE: {{user_objective}}
CURRENT STATE / COMPLETED STEPS: {{execution_state_json}}
TASK: Determine the next atomic action required to complete the objective with zero
redundancy.
OUTPUT STRICT JSON:
{
  "status": "IN_PROGRESS | READY_FOR_CRITIC | COMPLETED",
  "next_agent": "RESEARCHER | COPYWRITER | DATA_ANALYST | CRITIC | NONE",
  "subtask_instruction": "Exact, self-contained prompt and context variables to pass to
the selected agent",
  "success_criteria": "Exact check the output must pass before moving to the next step"
}
RULE: Never allow more than 2 revision loops between Specialist and CRITIC. If
iteration_count >= 2, finalize best output.
```

EXPECTED DELIVERABLE: Deterministic state-machine router payload for LangGraph or n8n loop workflows.

OPTIMIZATION LEVER: Always enforce a hard `max_iterations = 3` counter in your workflow loop to prevent infinite API token burn.

PROMPT #A502 **System Prompt: The Adversarial Critic & Quality Gatekeeper Node** OpenAI o3-mini / Claude 3.5 Sonnet

```
ROLE: You are a Ruthless Quality Assurance Critic Node in an automated pipeline.
ORIGINAL SPECIFICATION: {{task_spec}}
CANDIDATE DRAFT FROM WORKER AGENT: {{worker_output}}
TASK: Audit the draft against these 5 mandatory invariants:
1. Fact Grounding: Are all numbers/claims backed by the provided input context (zero
fabricated stats)?
2. Anti-Cliche Compliance: Does it contain banned AI words (delve, tapestry, unlock, game-
changer, realm)?
3. Completeness: Were all requested sections, tables, and variables fulfilled without
truncation?
4. Specificity: Are action items concrete enough to execute today without guessing?
5. Formatting: Does it strictly match the required Markdown or JSON schema?
OUTPUT STRICT JSON:
{
  "score": 0 to 100,
  "passed": true or false (true only if score >= 85),
  "violations": ["Exact quote of weak line -> Exact instruction on how to fix it"],
  "revised_instructions_for_worker": "Consolidated fix prompt if passed == false"
}
```

EXPECTED DELIVERABLE: Automated quality gate that catches hallucinations and generic fluff before a human ever sees the draft.

OPTIMIZATION LEVER: Adding a single $0.005 Critic pass improves blind human preference ratings of automated outputs from 45% to 88%.

MODULE 06 • IMPLEMENTATION BLUEPRINT

Production Reliability, Token FinOps & Guardrail Engineering

How to cut your OpenAI/Anthropic API bill by 75% while guaranteeing 99.9% workflow uptime with idempotency, retries, and dead-letter queues.

COST REDUCTION: -75% Token Spend | RELIABILITY: Enterprise Fault-Tolerant

Failure / Cost Vector	Naive Amateur Setup	Production-Grade Engineering Pattern	Measured Impact
Webhook Duplicate Fires	Stripe/Typeform retries webhook -> charges or emails client twice.	Store `event_id` in Supabase/Redis idempotency table before execution.	0% Duplicate Embarrassment
LLM Rate Limits (HTTP 429)	Workflow crashes mid-run and loses lead data.	Configure HTTP node with 3x Exponential Backoff (2s, 8s, 32s) + Fallback Model (GPT-4o -> Claude).	99.95% Completion Rate
Malformed JSON Crash	LLM adds "Here is your JSON:" preamble -> `JSON.parse()` throws syntax error.	Enforce Structured Outputs (`json_schema`) or strip markdown fences via Regex before parsing.	100% Parse Stability
Runaway API Token Bills	Using flagship Claude Sonnet / GPT-4o for simple routing & extraction.	Route 80% of classification/extraction to GPT-4o-mini / Haiku; enable Anthropic Prompt Caching.	-78% Monthly API Cost
Silent Background Failures	Broken OAuth token stops leads for 2 weeks unnoticed.	Attach a Global Error Trigger Workflow that posts failed payload + replay link to Slack `#ops-alerts`.	Instant 60s MTTR Recovery

PROMPT #A601 **System Prompt: Automated Workflow Error Triage & Self-Healing Diagnostic Bot** GPT-4o-mini (Global Error Workflow)

```
ROLE: You are an Automated DevOps SRE Agent triggered by an n8n / Make.com Global Error Handler.
FAILED WORKFLOW NAME: {{workflow_name}} | FAILED NODE: {{node_name}}
ERROR MESSAGE & HTTP CODE: {{error_message}} | INPUT PAYLOAD SNIPPET: {{input_json}}
TASK: Diagnose the failure and generate a clean Slack alert block for the founder:
1. Severity Level: [CRITICAL REVENUE IMPACT / WARNING / TRANSIENT RETRY].
2. Root Cause in Plain English (e.g., "HubSpot OAuth token expired", "Prospect website timed out after 10s", "OpenAI 429 quota exceeded").
3. Immediate Fix Command: Exact step to resolve and replay the payload in under 2 minutes.
```

EXPECTED DELIVERABLE: Instant plain-English Slack alert whenever any background automation fails.

OPTIMIZATION LEVER: Wire this into n8n's "Error Trigger" node so one single error workflow monitors all 20+ of your active automations.

The 12 Core Solopreneur Automations Ranked by ROI & Build Time

Deploy these workflows in exact priority order over 4 weekends to reclaim 25+ hours per week.

Priority	Automation Blueprint	Build Time	Hours Saved/Mo	Direct Revenue / Cost Impact
#01	Sub-60s Inbound Lead Enrichment & Reply Bot	2.5 Hours	18 hrs/mo	+200% to +390% Lead-to-Call Conversion
#02	Stripe -> Notion Client Portal & Kickoff Provisioner	2.0 Hours	12 hrs/mo	Zero buyer remorse, instant onboarding
#03	Failed Payment (Dunning) Multi-Channel Recovery	1.5 Hours	6 hrs/mo	Recovers 60%+ of failed subscription cards
#04	Omnichannel Content Engine (1 Voice Note -> 12 Posts)	3.0 Hours	32 hrs/mo	Consistent daily authority across 4 channels
#05	Sales Call Transcript -> MEDDIC Audit + Proposal Draft	2.0 Hours	15 hrs/mo	Proposals sent in 15 mins while intent is hot
#06	Clay/Apollo Signal-Based Outbound Icebreaker Pipeline	3.0 Hours	25 hrs/mo	8-15% positive reply rate on autopilot
#07	Customer Support RAG Triage & Draft Responder	2.5 Hours	20 hrs/mo	Resolves 75% of Tier-1 tickets in 30 seconds
#08	Weekly Competitor Pricing & Ad Library Intelligence Swarm	2.5 Hours	10 hrs/mo	Early detection of market shifts & ad hooks
#09	Automated Testimonial & Case Study Collector (NPS > 9)	1.5 Hours	8 hrs/mo	Continuous stream of fresh social proof
#10	Friday CFO Cash-Flow, Stripe & Ad-Spend Slack Digest	1.5 Hours	6 hrs/mo	Real-time unit economics & burn visibility

[PRODUCTION GATE] PRE-FLIGHT PRODUCTION LAUNCH CHECKLIST (RUN BEFORE ACTIVATING ANY WORKFLOW)

[ ] 1. Idempotency check enabled on webhook entry point (no duplicate runs on network retry).
[ ] 2. All LLM nodes enforce strict JSON Schema (`response_format: json_object`) with temperature <= 0.3 for data tasks.
[ ] 3. Hard loop counter (`max_iterations <= 3`) configured on all multi-agent critic loops.
[ ] 4. HTTP timeout set to 30,000ms with 3x exponential backoff on 429/502/503 status codes.
[ ] 5. Connected to Global Slack Error Handler with payload logging enabled.

The 12 Core Solopreneur Automations Ranked by ROI & Build Time

Deploy these workflows in exact priority order over 4 weekends to reclaim 25+ hours per week.

Priority	Automation Blueprint	Build Time	Hours Saved/Mo	Direct Revenue / Cost Impact
#01	Sub-60s Inbound Lead Enrichment & Reply Bot	2.5 Hours	18 hrs/mo	+200% to +390% Lead-to-Call Conversion
#02	Stripe -> Notion Client Portal & Kickoff Provisioner	2.0 Hours	12 hrs/mo	Zero buyer remorse, instant onboarding
#03	Failed Payment (Dunning) Multi-Channel Recovery	1.5 Hours	6 hrs/mo	Recovers 60%+ of failed subscription cards
#04	Omnichannel Content Engine (1 Voice Note -> 12 Posts)	3.0 Hours	32 hrs/mo	Consistent daily authority across 4 channels
#05	Sales Call Transcript -> MEDDIC Audit + Proposal Draft	2.0 Hours	15 hrs/mo	Proposals sent in 15 mins while intent is hot
#06	Clay/Apollo Signal-Based Outbound Icebreaker Pipeline	3.0 Hours	25 hrs/mo	8-15% positive reply rate on autopilot
#07	Customer Support RAG Triage & Draft Responder	2.5 Hours	20 hrs/mo	Resolves 75% of Tier-1 tickets in 30 seconds
#08	Weekly Competitor Pricing & Ad Library Intelligence Swarm	2.5 Hours	10 hrs/mo	Early detection of market shifts & ad hooks
#09	Automated Testimonial & Case Study Collector (NPS > 9)	1.5 Hours	8 hrs/mo	Continuous stream of fresh social proof
#10	Friday CFO Cash-Flow, Stripe & Ad-Spend Slack Digest	1.5 Hours	6 hrs/mo	Real-time unit economics & burn visibility

[PRODUCTION GATE] PRE-FLIGHT PRODUCTION LAUNCH CHECKLIST (RUN BEFORE ACTIVATING ANY WORKFLOW)

[ ] 1. Idempotency check enabled on webhook entry point (no duplicate runs on network retry).
[ ] 2. All LLM nodes enforce strict JSON Schema (`response_format: json_object`) with temperature <= 0.3 for data tasks.
[ ] 3. Hard loop counter (`max_iterations <= 3`) configured on all multi-agent critic loops.
[ ] 4. HTTP timeout set to 30,000ms with 3x exponential backoff on 429/502/503 status codes.
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